

4 Accessing location-based services

4.1 Overview

One of the most popular features on current mobile devices is GPS capability. Besides GPS, your phone can also use alternatives like cell tower triangulation and proximity to public Wi-Fi hotspots to identify your location.

While the most prevalent uses of location-based services are for mapping and getting directions, there are other things you can do if you know your location. You can set up a dynamic chat application based on physical location, to chat with people near your location like Buzz on Google Maps. You can also automatically geo-tag posts/pictures to Twitter or similar services.

Android devices may have one or more of these services available to them. You, as a developer, can ask the device for your location, plus details on which providers are available. There are even ways for you to simulate your location in the emulator, for use in testing your location-enabled applications.

4.2 Location providers

Android devices can access several different means of determining your location. The SDK has abstracted all this out into a set of `LocationProvider` objects. Your Android environment will have zero or more `LocationProvider` instances: one for each distinct locating service that is available on the device. Providers know not only your location, but also are aware of their own characteristics, in terms of accuracy, cost, and so on.

You, as a developer, will use a `LocationManager`, which holds the `LocationProvider` set, to figure out which `LocationProvider` is right for your particular circumstance. You'll also need a permission in your application, or the various location APIs will fail due to a security violation. Depending on which location providers you wish to use, you may need `ACCESS_COARSE_LOCATION`, `ACCESS_FINE_LOCATION`, or both.

4.3 Finding yourself

The most obvious thing to do with a location service is to figure out where you are right now. To determine your current location, first you need to get a `LocationManager` call `getSystemService(LOCATION_SERVICE)` from your activity or service and cast it as a `LocationManager`.

The next step is to get the name of the `LocationProvider` you want to use. Here, you have two main options: